



Earth System

Earth System Science

Chapter 2: THE COMPONENTS OF THE EARTH SYSTEM



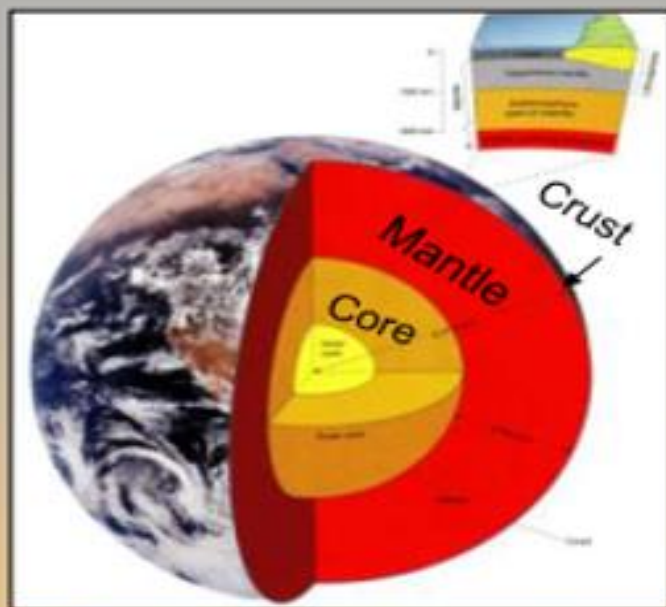
Objective

Geosphere

- *To Define The earth's layered structure*
- *To Know The plate tectonic*
- *To See The geologic features*
- *To explain the rock cycle*

What Is the Geosphere?

- The geosphere includes the solid part of Earth rocks, minerals, landforms, and the processes that shape them (plate tectonics, volcanism, erosion).
- Energy sources:
 - Internal heat from radioactive decay and residual heat of Earth's formation.
 - External energy from the Sun driving surface processes (weathering and erosion). These energy flows power plate tectonics, volcanism, and mountain building.



The layers of the earth.

Modified after Plummer/McGeary, 7th ed., pg. 14

The Earth is made up of three main layers:

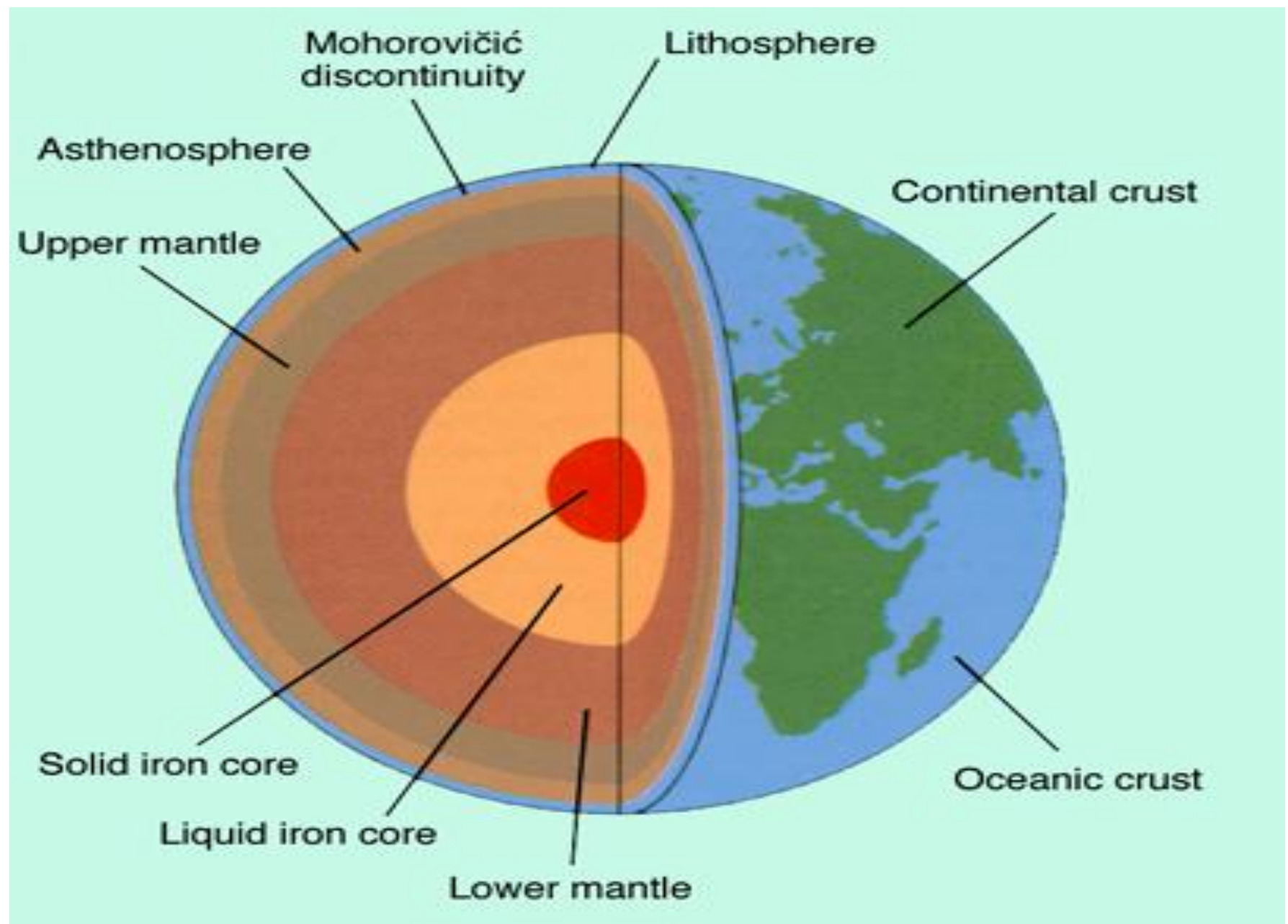
1. The Core is at the center of the Earth. It is divided into an inner and outer core.
2. The Mantle is the layer surrounding the core. The upper mantle is partially molten and called the asthenosphere.
3. The Crust, or lithosphere, is the rigid outer-most layer. Thick continental crust underlies continents, and thin, very dense oceanic crust underlies oceans.

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	<u>Inner Core</u>	<u>Outer Core</u>	<u>Mantle</u>	<u>Crust</u>
Thickness	1,216 km	2,270 km	2,900 km	Continental 35-90 km Oceanic 7-8 km
Physical Properties	Solid Iron; extremely dense (17 g/cm ³)	Molten Iron, very dense (12 g/cm ³)	Made mostly of silicates of magnesium and iron; moderately dense. Behaves like melted plastic in upper-most section (5.5 g/cm ³)	Made of silicate rocks and oxides; slightly dense; rigid. (2.67-3.3 g/cm ³)
Percentage of Earths' Mass	30%		65%	5%

Activate Window

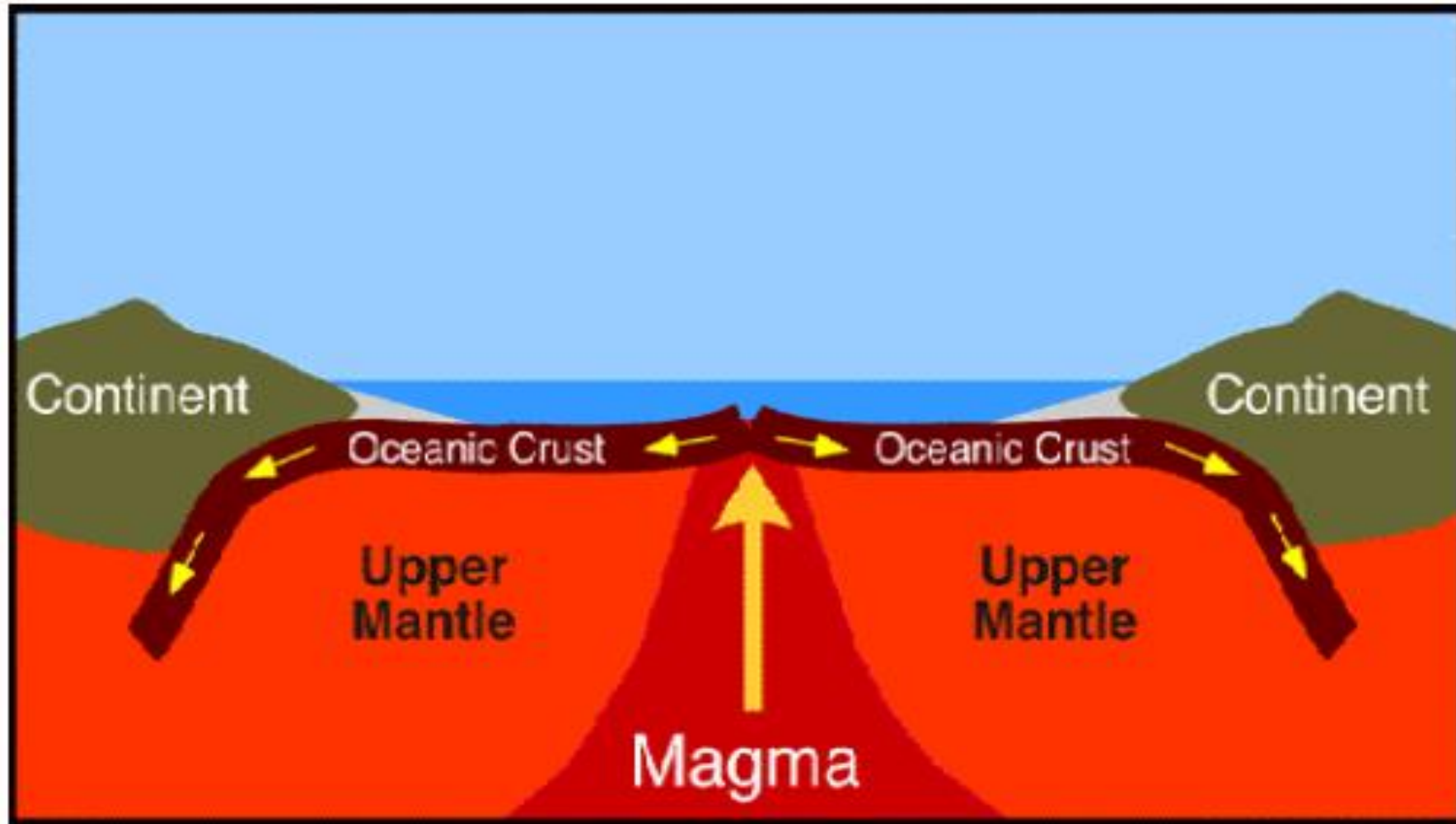


The Plate Tectonic

Crust (lithosphere) Plate Movement

The crust is broken into plates that move.

- Lithosphere is divided into plates that slide around on top of the slowly moving mesosphere (tectonic plates)
- It occurred after the break-up of Pangaea, and since then the continents have drifted apart



THEORY OF PLATE TECTONICS

The theory of plate tectonics originated in the late 1960's by combining two pre-existing theories:

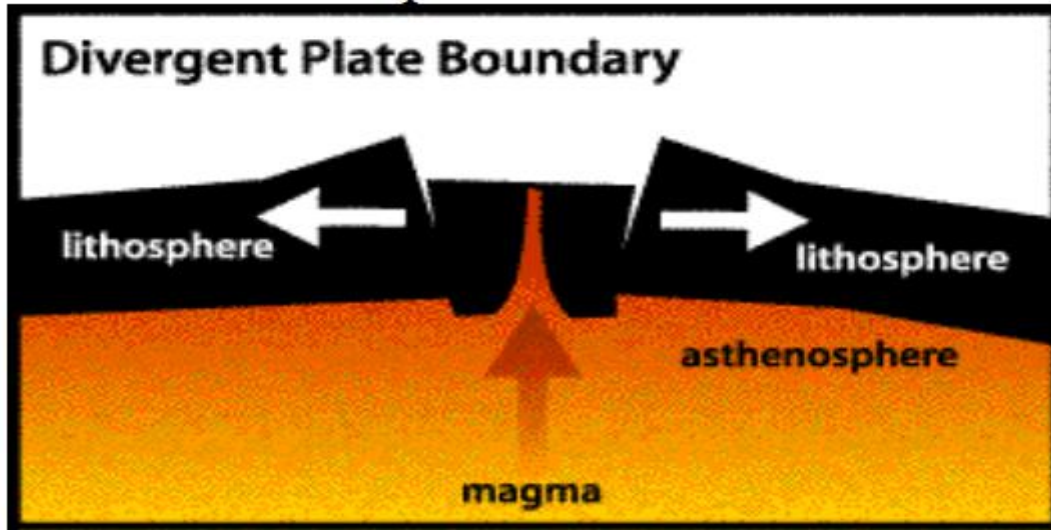
- 1. Continental Drift – The idea that continents move freely over the Earth's surface, changing their positions relative to one another**
- 2. Sea-Floor Spreading – A hypothesis that the sea floor forms at the crest of the mid-oceanic ridge, then moves horizontally away from the ridge crest toward an oceanic trench**

How does continental drift work?

- Plates collide = convergent boundary
- Plates separate = divergent boundary
- Plates slide past one another horizontally = transform boundary
- Subduction zone = one plate slides under another

Divergent Boundary

- Plates separate



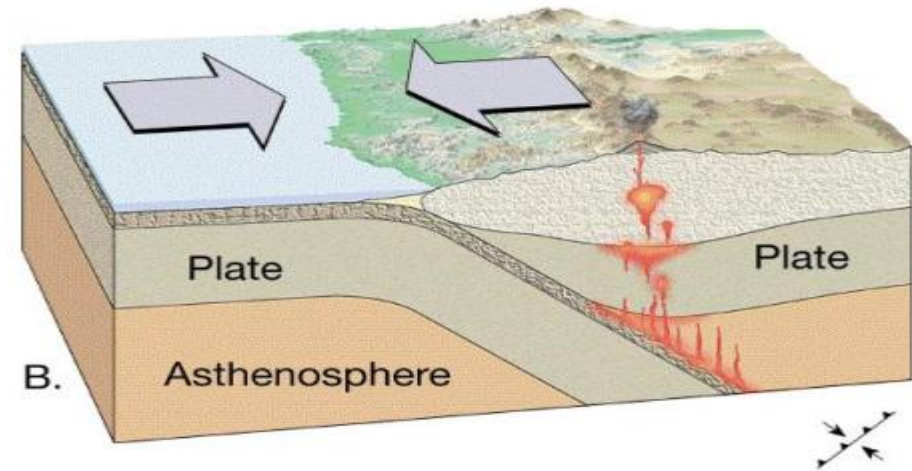
Transform Boundary



Transform Fault

Plates slide past one another horizontally

Convergent Boundary/subduction zone

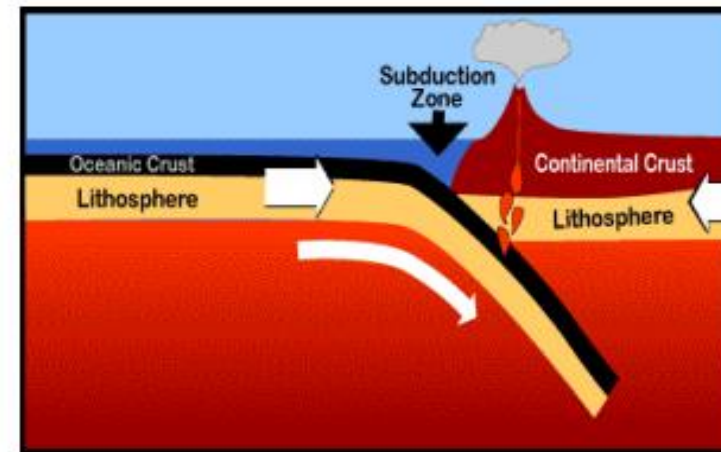


What happens when earthquakes happen in the ocean?

TSUNAMI wave that brings lots of water to the continents and causes major flooding

Subduction Zone

- one plate slides under another



Oceanic trench forms at the subduction zone. Volcanoes may form along this type of boundary.

What are Tectonic Plates? (continued)

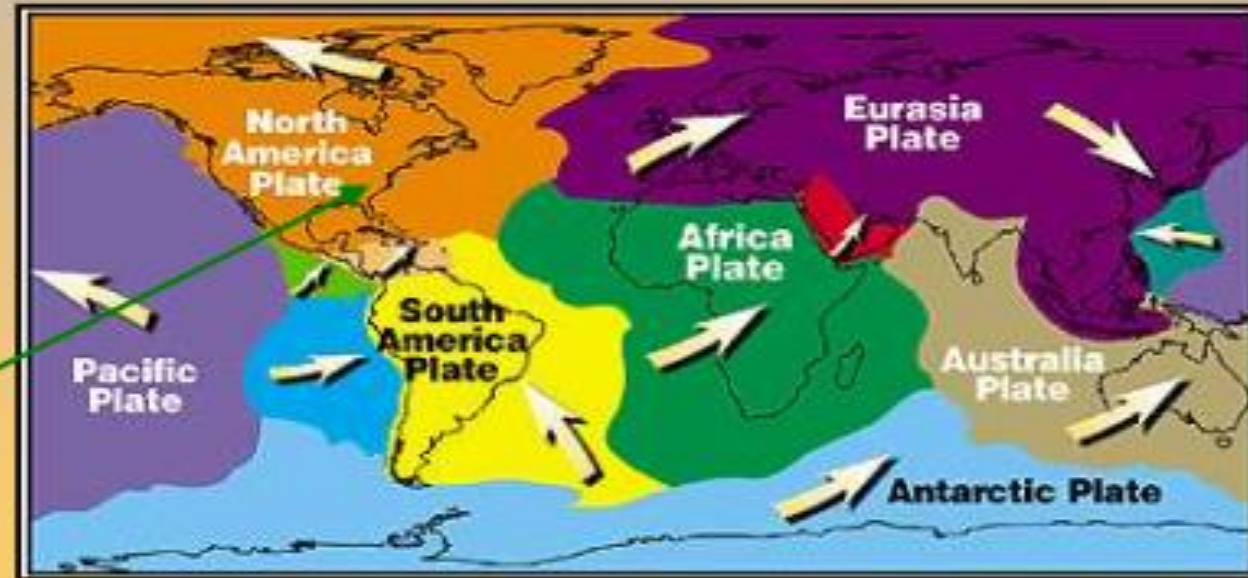
❖ The Earth's crust consists of about a dozen large slabs of rock, or **PLATES**, that the continents and oceans rest on. These tectonic plates can move centimeters per year— about as fast as your fingernails grow up to 15cm/yr in some places.

❖ Tectonic plates are also called **lithospheric plates** because the crust and the upper-most mantle make up a sub-layer of the earth called the lithosphere. The plates can move about because the uppermost mantle, or the asthenosphere, is partially molten and possesses a physical property called plasticity, allowing the strong, rigid plates of the crust to move over the weaker, softer asthenosphere.

Plates and relative plate motion.

Modified after NOAA

South Carolina is located on the North American plate



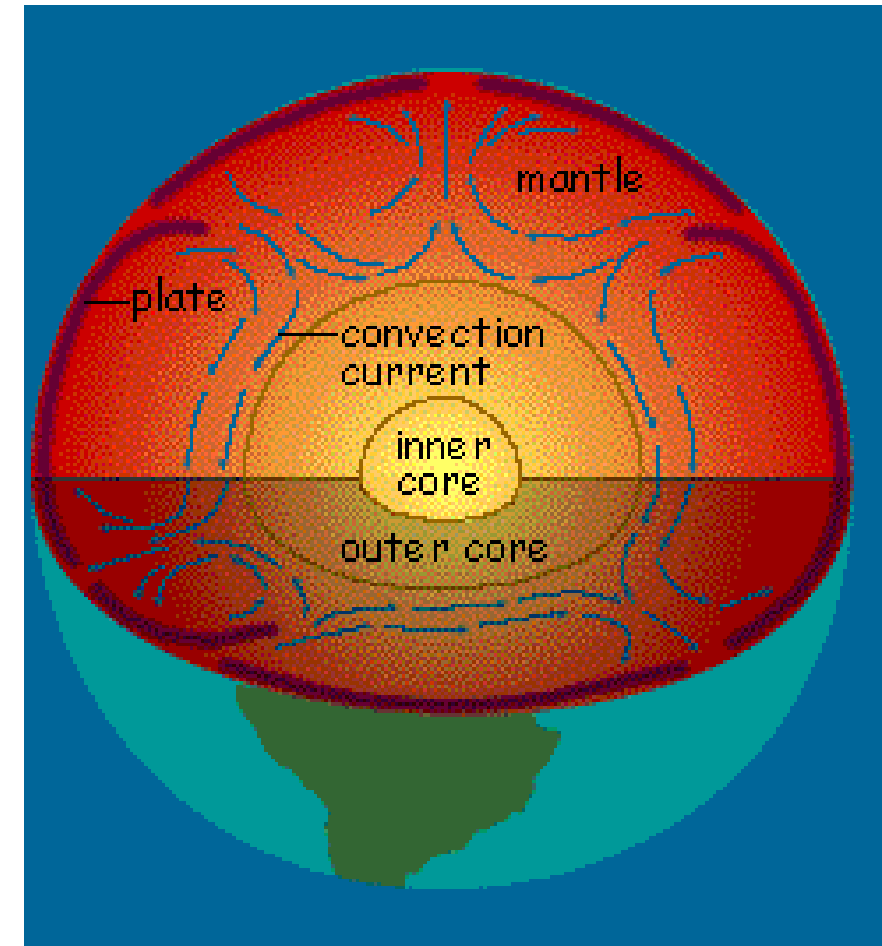
❖ The word **TECTONICS** is of Greek origin and it means “to build.” The word “tectonism” refers to the deformation of the lithosphere. This deformation most notably includes mountain building.

What are Tectonic Plates? (continued)

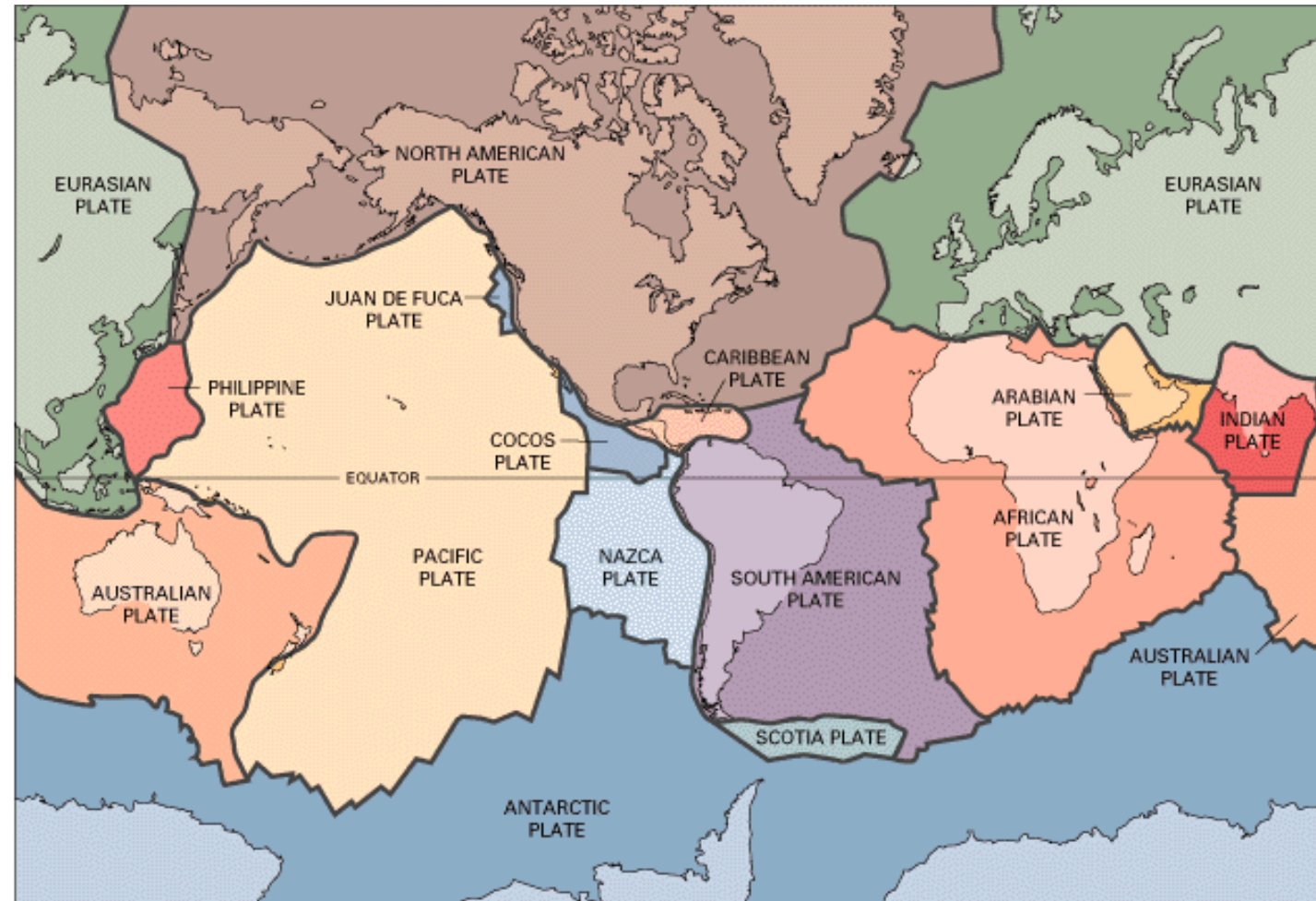
❖ Tectonic plates, or lithospheric plates, are constantly moving, being created, and consumed simultaneously. The motion sometimes results in earthquakes, volcanoes, and mountain ranges at the plate boundaries.

What Drives Plate Tectonics?

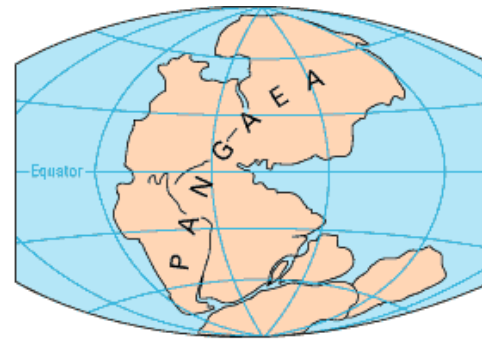
- The slow movement of hot, softened mantle lies below rigid plates.
- The hot, softened rock in the mantle moves in a circular manner in a convection flow – the heated, molten rock rises to the surface, spreads, and begins to cool, and then sinks back down to be reheated and rises again.



Earth's Plates



Movement of the Plates Over Time



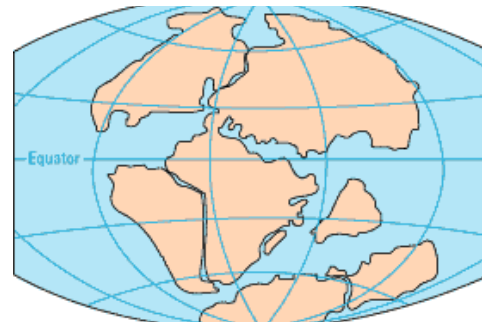
Permian

248 million years ago



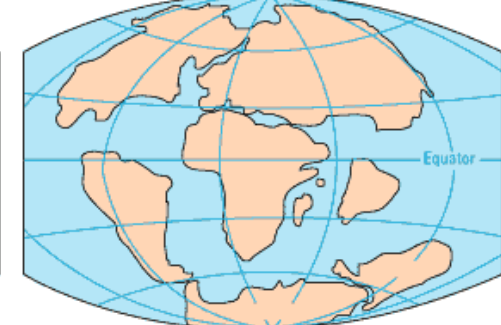
Triassic

206 million years ago



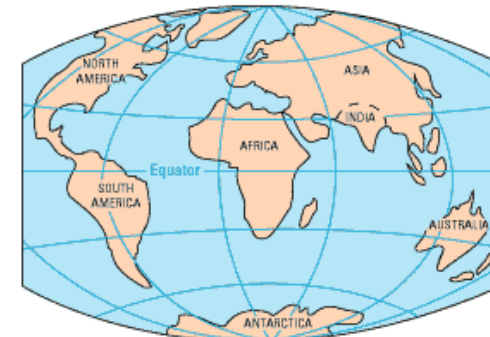
Jurassic

206 million years ago



Cretaceous

65 million years ago



Present Day

Consequences of Plate Tectonics



The San Andreas Fault – a transform fault



Aerial view of the area around Thingvellir, Iceland, showing a fissure zone (in shadow) that is an on-land exposure of the Mid-Atlantic Ridge. Right of the fissure, the North American Plate is pulling westward away from the Eurasian Plate (left of fissure).

Consequences of Plate Tectonics



The Aleutian Islands, an island arc



The 1980 eruption of Mount Saint Helens

Consequences of Plate Tectonics



Helicopter view (in February 1994) of the active lava lake within the summit crater of 'Erta 'Ale (Ethiopia), one of the active volcanoes in the East African Rift Zone.



The convergence of the Nazca and South American Plates has deformed and pushed up limestone strata to form towering peaks of the Andes, as seen here in the Pachapaqui mining area in Peru.

The Geologic Features

Volcanoes

- Among the most striking geological features are **volcanoes**.
- These are openings where magma comes out from inside the Earth to the surface.
- They are often recognized as tall mountain peaks, but there are many different kinds of volcanoes, some of which are much smaller and flatter than the ones we imagine with dramatic eruptions.
- Magma itself comes from the **mantle**, the layer of Earth just below the crust, and it's a strange material that is partially gaseous, partially liquid, and partially solid.

The role of Volcanoes in the Earth system

Atmosphere

- *Gas Emissions*: Volcanoes release gases like carbon dioxide (CO₂), sulfur dioxide (SO₂), water vapor, and trace gases.
- *Aerosols and Ash*: Volcanic ash and aerosols can temporarily reduce global temperatures by blocking sunlight but may also contribute to longer-term warming by adding greenhouse gases.
- *Ozone Depletion*: Halogen compounds released during eruptions can deplete stratospheric ozone.

Hydrosphere

- *Water Chemistry*: Volcanic activity releases minerals and chemicals into water systems, altering pH and nutrient levels in oceans, lakes, and rivers.

Lithosphere

- Crust Formation

Biosphere

- Ecosystem Disruption

Canyons

- **Canyons** are another unique geological feature on Earth. They are valleys on the surface of Earth that are both deep and narrow.
- Rivers are a major factor in how canyons form, as strong moving currents erode and carve out the dry landscape.
- The rock layers that are visible in canyon walls tell stories about the history of the landscape and provide scientists with information about mineral deposits, geological events, and more.
- The Grand Canyon, arguably the most famous canyon in the world, was carved out over millions of years by the Colorado River.
- Some canyons may start from a small spring, periodic bursts of water, or even currents on the ocean floor. Some canyons may be only a few feet wide but hundreds of feet deep, making them beautiful wonders to behold.

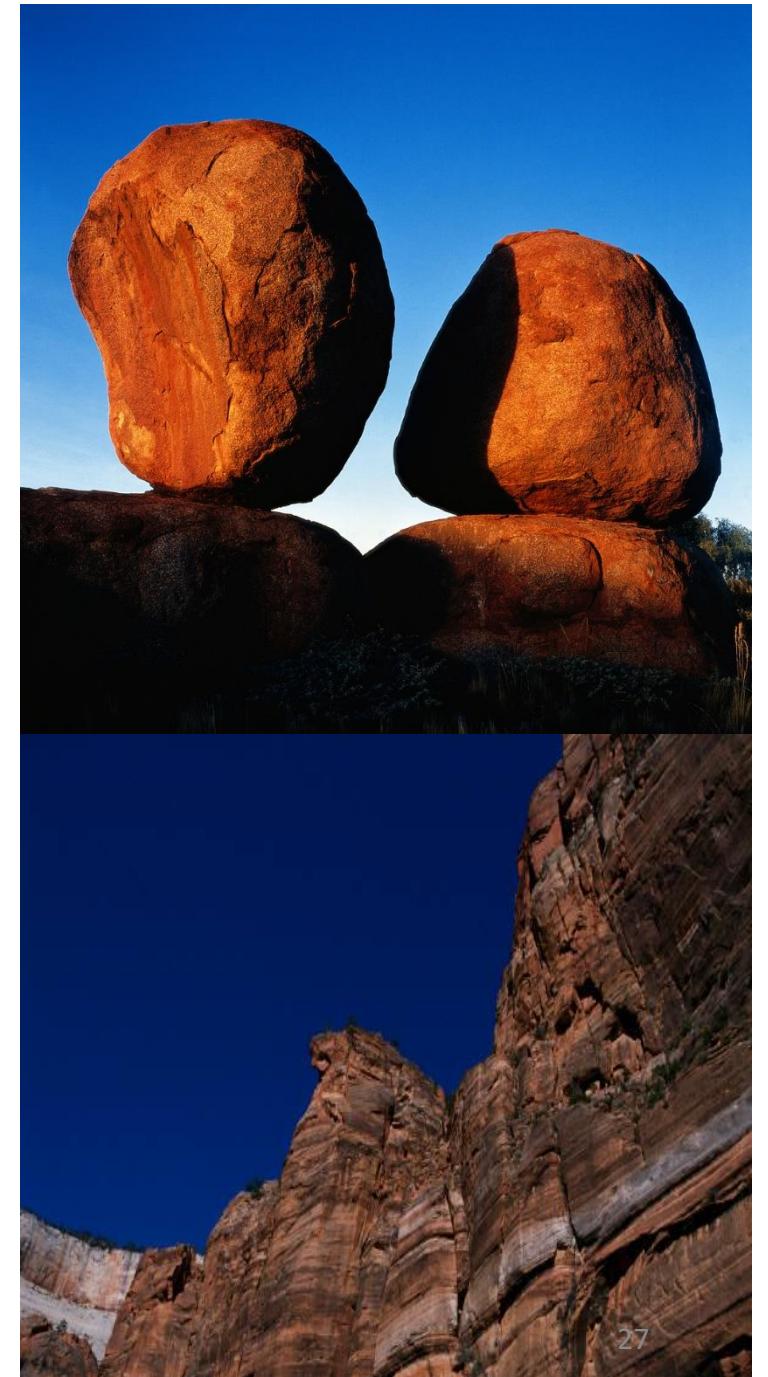


Canyons influence the Earth system by

- Lithosphere:** Exposing geological history and shaping landforms through erosion and tectonic activity.
- Hydrosphere:** Regulating water flow, transporting sediments, and connecting surface and groundwater systems.
- Biosphere:** Hosting diverse ecosystems and acting as habitats for unique flora and fauna.
- Atmosphere:** Creating microclimates, affecting wind and weather patterns, and influencing erosion-related dust release.

The Rock Cycle

- Rocks are aggregates of 2 or more minerals.
- Petrology is the study of rocks.
- There are three classifications of rocks: igneous, sedimentary, and metamorphic.



• **IGNEOUS**

- **Forms from melted rock that has cooled & hardened**



- **Made from lava and magma.**

Formation of Igneous Rocks

- When some volcanoes erupt, they eject a flow of molten rock material.
- Molten rock material, called magma, flows when it is hot and becomes solid when it cools.
- When hot magma cools and hardens, it forms **igneous rock**.



• METAMORPHIC

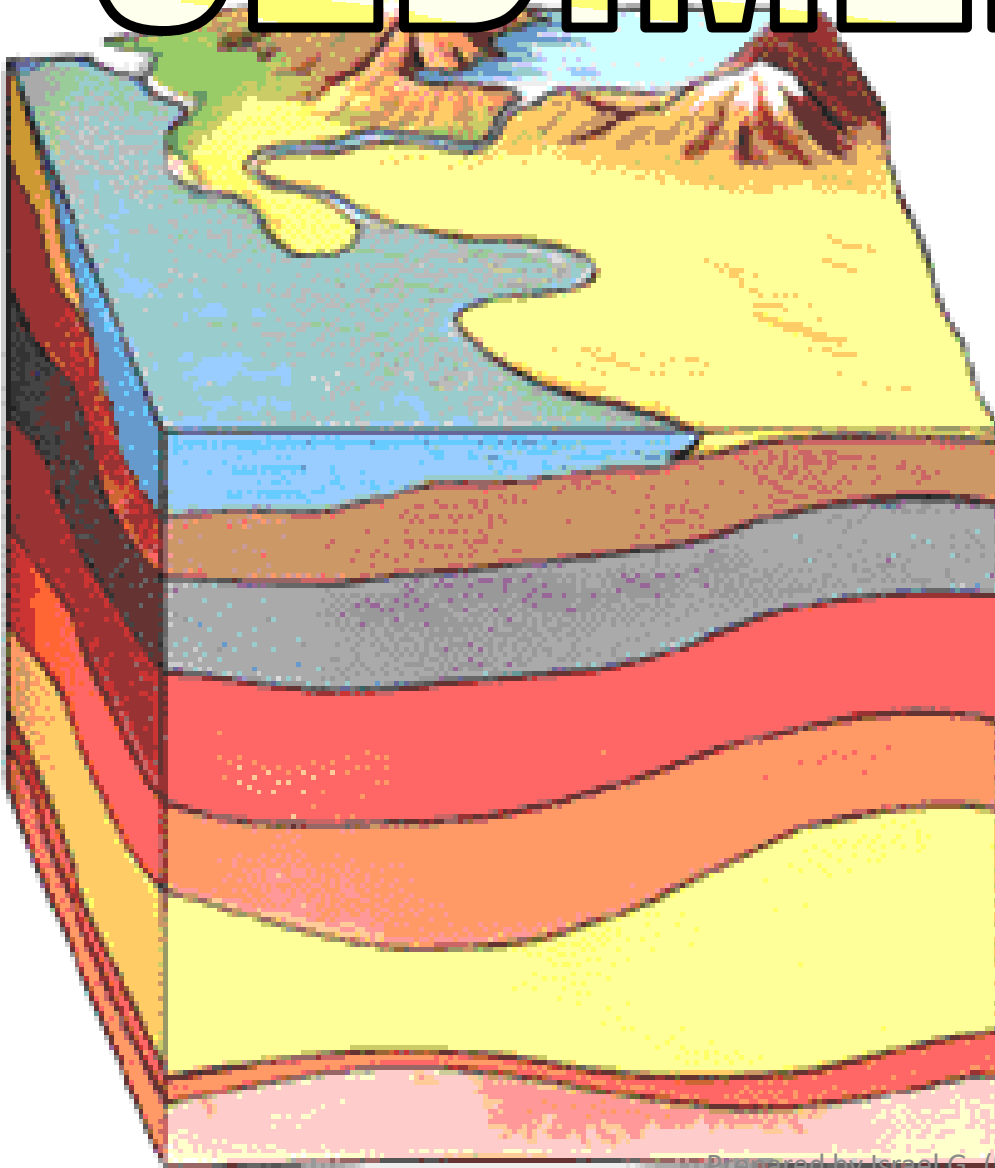
- Rocks formed by heat and pressure.



Formation of Metamorphic Rocks

- Rocks that have changed because of changes in temperature and pressure (heat & pressure) or the presence of hot watery fluids are called **metamorphic rocks**.

◦ SEDIMENTARY



◦ SEDIMENTARY

- layers of sediment cemented and compacted together



- Caused by weathering and erosion.

Formation of Sedimentary Rocks

- **Sediments** are loose materials such as rock fragments, mineral grains, and bits of shell that have been moved by wind, water, ice, or gravity.
- Sediments come from already-existing rocks that are weathered and eroded.
- **Sedimentary rock** forms when sediments are pressed and cemented together, or when minerals form from solutions.

Weathering and Erosion

- When rock is exposed to air, water, or ice, it breaks down chemically and mechanically.
- This process, which breaks rocks into smaller pieces, is called weathering.
- These pieces are classified by size.
- The movement of weathered material is called erosion.

SEDIMENTS

ARE MADE BY:



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- WEATHERING
- AND
- EROSION

• **WEATHERING**

- **to break down
or break apart
rock into
smaller pieces**

• WEATHERING

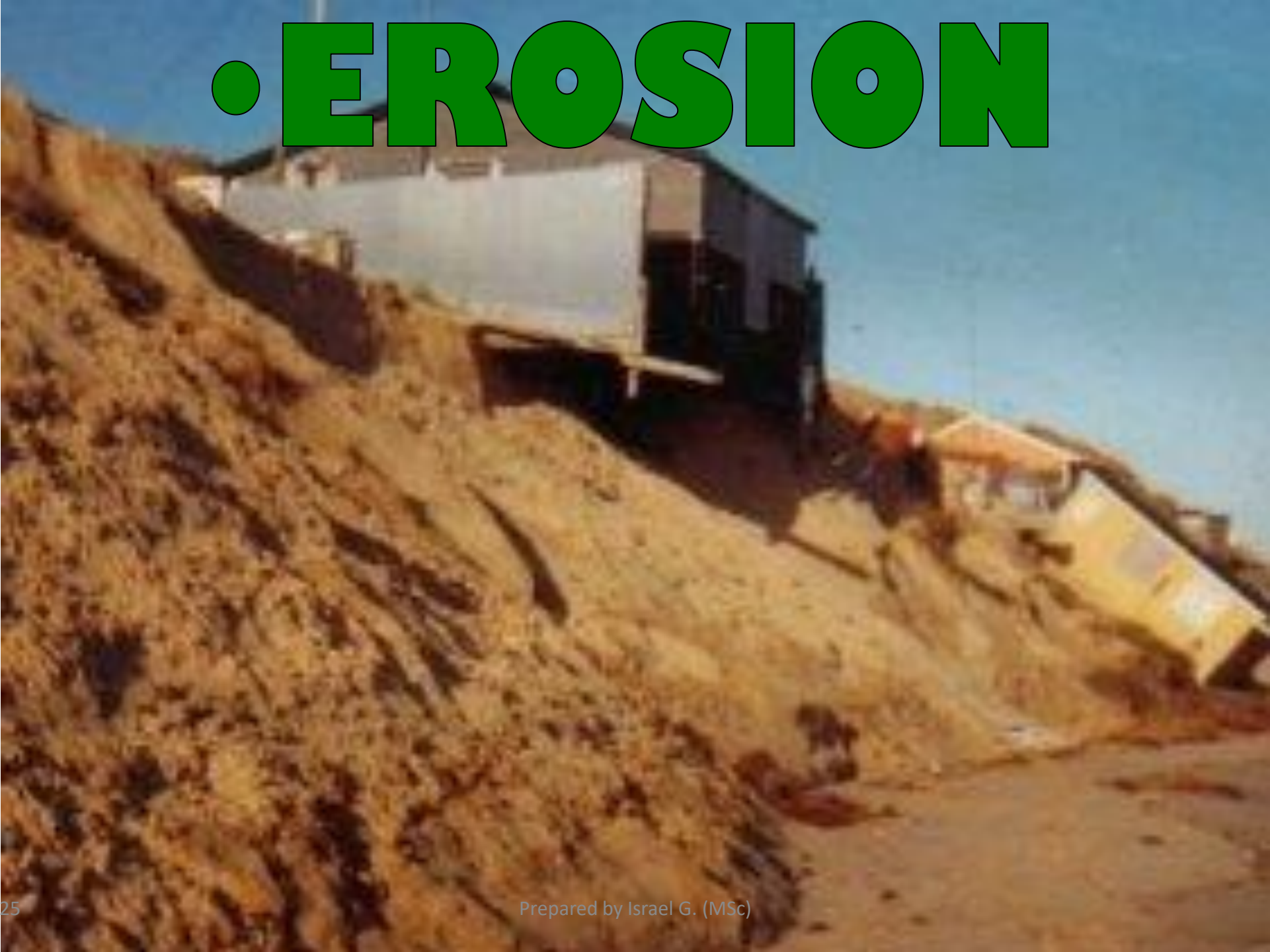


• EROSION

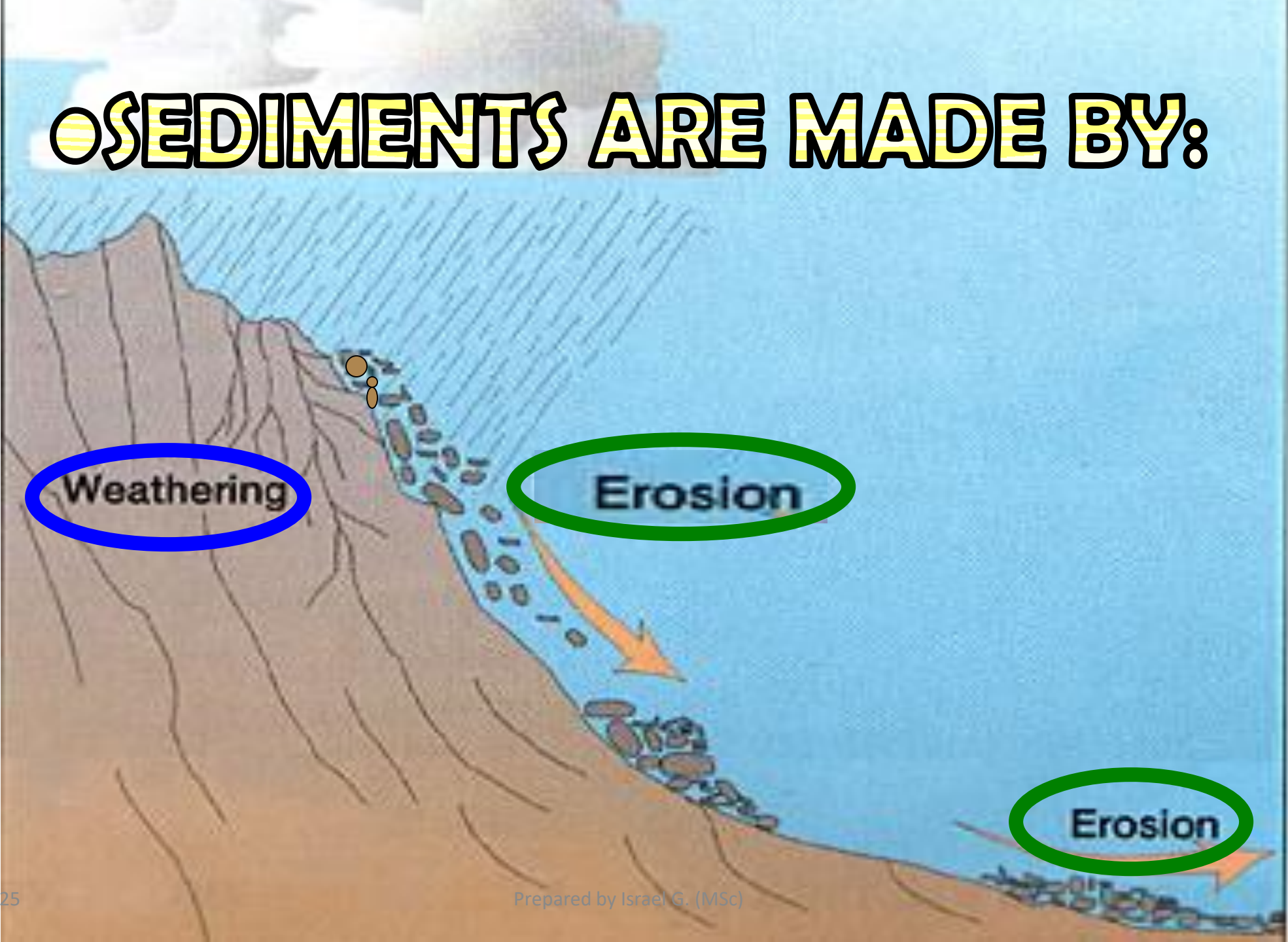


• to break down
and carry away
pieces of
sediment

• EROSION



SEDIMENTS ARE MADE BY:



The Rock Cycle

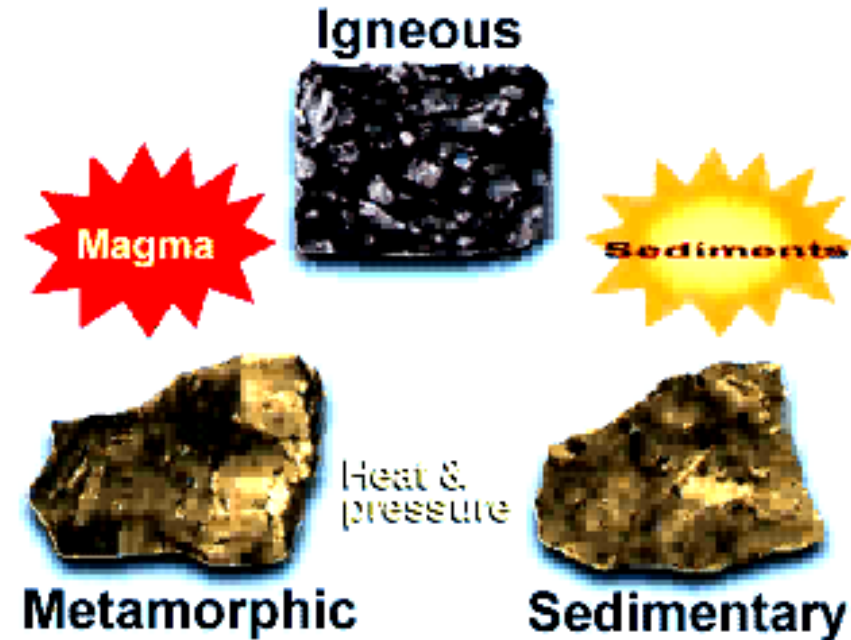
The Rock Cycle

- To show how rocks slowly change through time, scientists have created a model called the **rock cycle**.

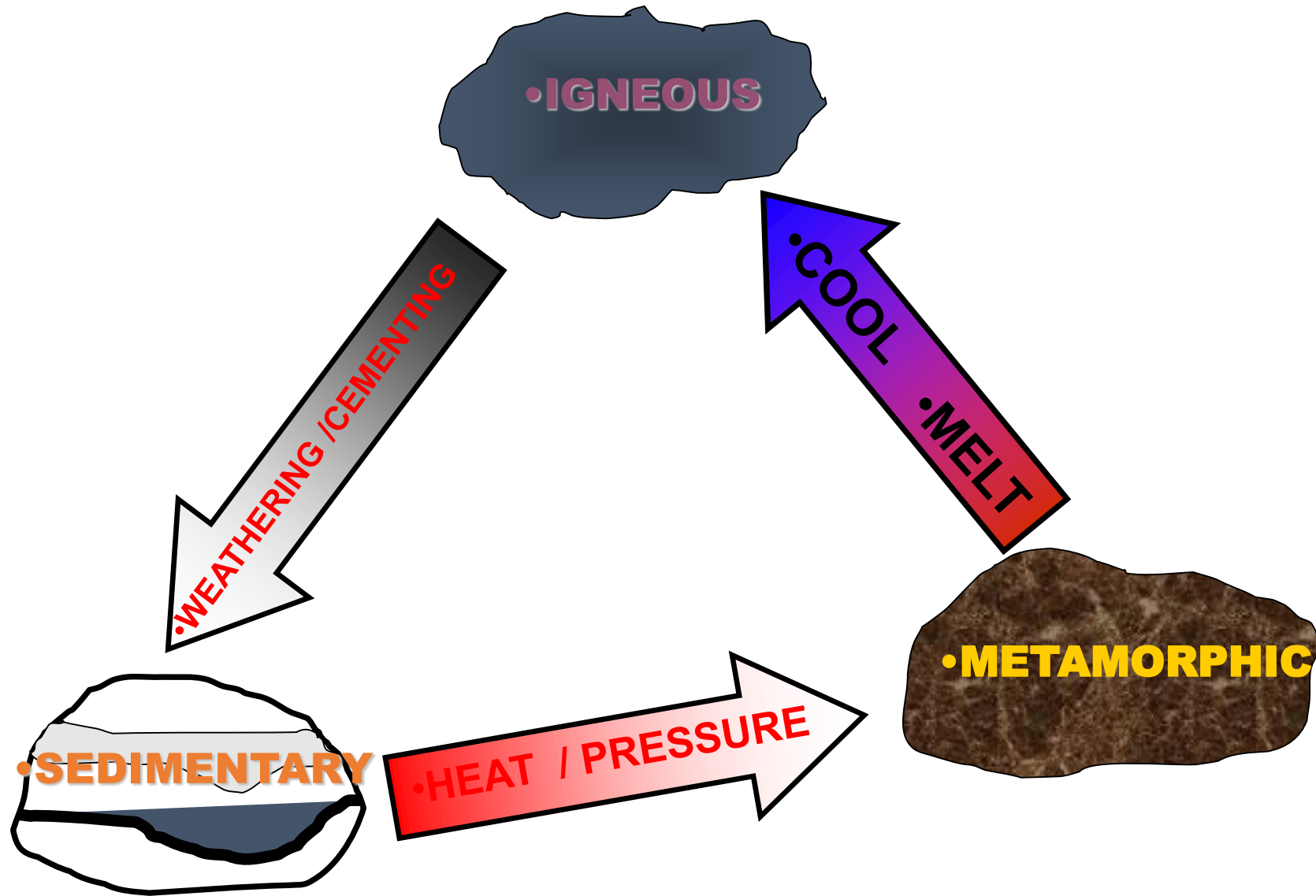
- It illustrates the processes that create and change rocks.

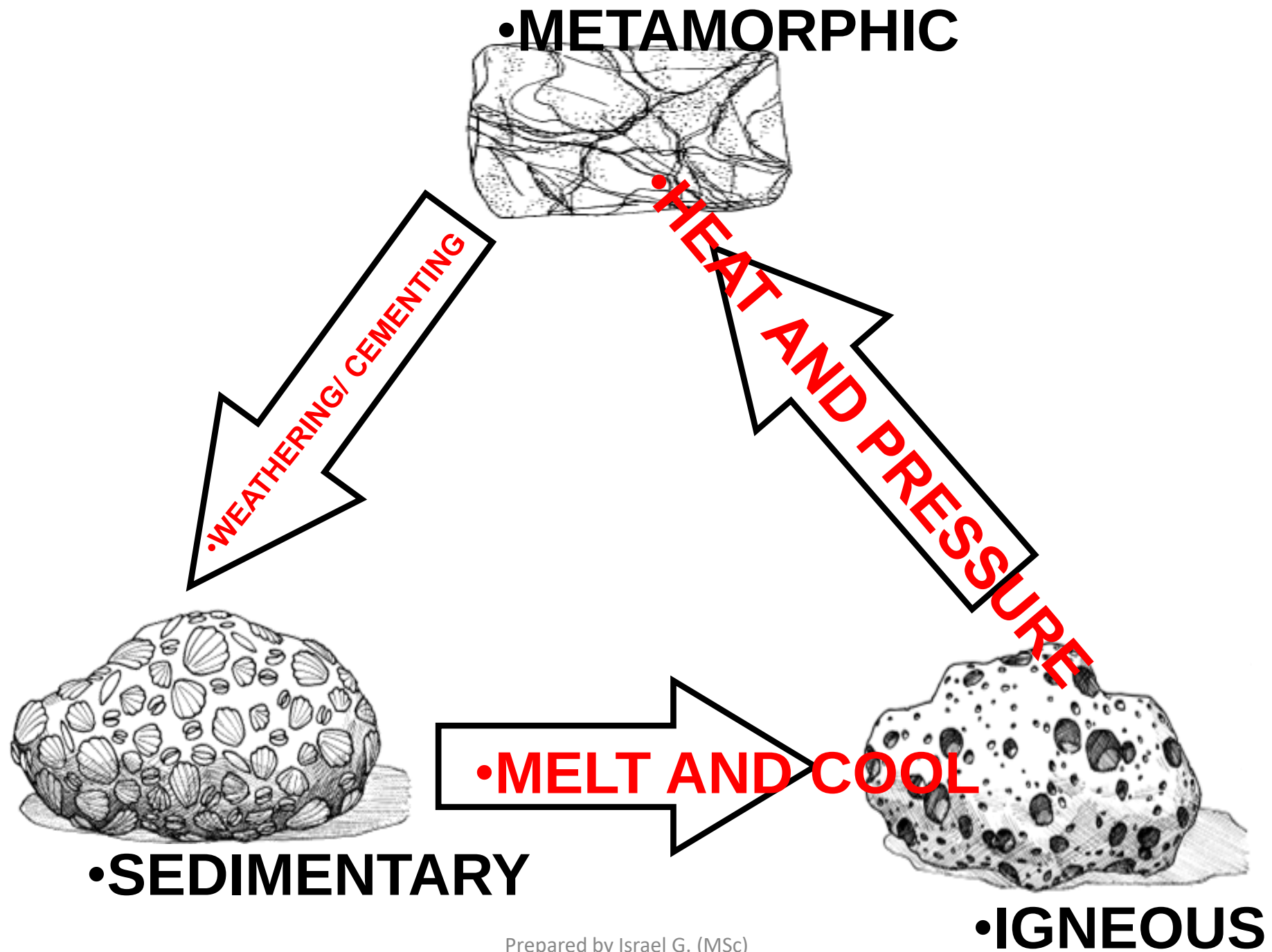
The Rock Cycle

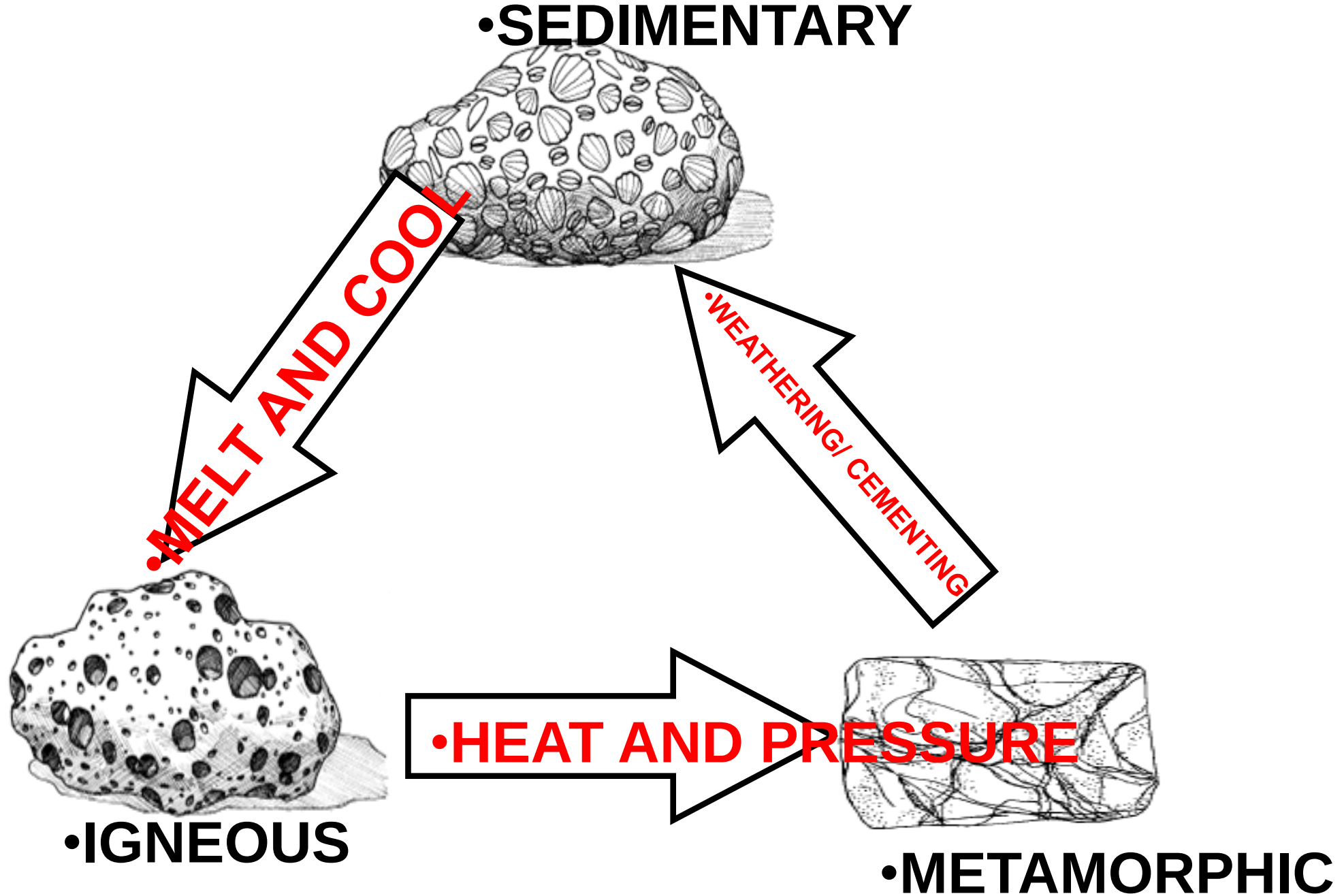
- The rock cycle shows the three types of rock—igneous, metamorphic, and sedimentary—and the processes that form them.



[Click image to view movie.](#)







The Rock Cycle

- Rocks change by many processes.
- For example, a sedimentary rock can change by heat and pressure to form a metamorphic rock.
- The metamorphic rock then can melt and later cool to form an igneous rock.

The Rock Cycle

- The igneous rock then could be broken into fragments by weathering and erode away.
- The fragments might later compact and cement together to form another sedimentary rock. Any given rock can change into any of the three major rock types. A rock even can transform into another rock of the same type.

....END

Thanks !